

Class Test 1

Date : 5 September 2018

Time : 6:00 – 7:30 pm

Marks available: 51

Full marks: 50

No calculators may be used.

1. Read the following theorem and the outline of its proof:

Theorem. Let $(a_k)_{k=0}^{\infty}$ be a sequence of complex numbers. If $\lim_{k \rightarrow \infty} \frac{|a_{k+1}|}{|a_k|}$ converges, then for each $z_0 \in \mathbb{C}$, the series $\sum_{k=0}^{\infty} a_k (z - z_0)^k$ has radius of convergence equal to $\lim_{k \rightarrow \infty} \frac{|a_k|}{|a_{k+1}|}$.

Proof. Let R be the radius of convergence of this power series and let $r = \lim_{k \rightarrow \infty} \frac{|a_k|}{|a_{k+1}|}$. Suppose that $0 < |z - z_0| < r$. We can then find an $\alpha < 1$ such that

$$|z - z_0| < \alpha \frac{|a_k|}{|a_{k+1}|} \quad (1)$$

for all large $k \in \mathbb{N}$. It follows that

$$\lim_{k \rightarrow \infty} \frac{|a_{k+1} (z - z_0)^{k+1}|}{|a_k (z - z_0)^k|} \leq \alpha. \quad (2)$$

Thus $\sum_{k=0}^{\infty} a_k (z - z_0)^k$ converges. From this, we can deduce that $r \leq R$. A similar argument shows that $R \leq r$, so $r = R$.

Now answer the following questions:

- (a) Explain carefully why it is possible to find an $\alpha < 1$ so that (1) holds.
- (b) Show how (2) follows from (1).
- (c) Immediately after (2), it is stated “Thus $\sum_{k=0}^{\infty} a_k (z - z_0)^k$ converges”. Explain carefully why this is the case.
- (d) How can we then deduce that $r \leq R$?
- (e) At the end of the proof, it is claimed that a similar argument shows that $R \leq r$. Write out the details of this argument. [2, 2, 3, 1, 5]

2. Let $f(z) = |z^2 - 4|^2$.

- (a) Is f differentiable at $z = 2$? (A “yes” or “no” answer will not suffice.)
- (b) Are the Cauchy–Riemann equations satisfied at $z = 2$? (Again, “yes” or “no” is not enough.)

3. Find the radius of convergence of each of the following power series.

(a) $\sum_{k=0}^{\infty} (3 + (-1)^k)^k z^k$

(b) $\sum_{k=1}^{\infty} \frac{k^k}{k!} (z + 1)^k$

[3, 4]

4. Write down a power series of the form $\sum_{k=0}^{\infty} a_k z^k$ for

(a) $\frac{2 + z}{1 - iz}$

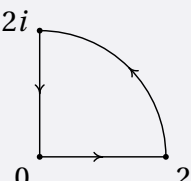
(b) $\frac{1}{4z^2 + 2z + 1}$

In each case, give the domain of convergence. (Give reasons for your answer.)

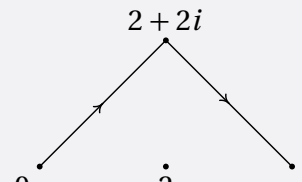
[6, 5]

5. Calculate each of the following integrals using the illustrated path:

(a) $\int_{\gamma} \bar{z} \, dz$



(b) $\int_{\gamma} \frac{z}{(z^2 - 4)^2} \, dz$



[10]

6. Find an upper bound on $\left| \int_{\gamma} \frac{1}{z^4 - 16} \, dz \right|$, where $\gamma(t) = 3e^{it}$, $0 \leq t \leq \pi$.

[5]

Class Test 1 solutions

- 1.(a) Since $|z - z_0| < r$, we can find an $\alpha < 1$ such that $|z - z_0| < \alpha r$. Since $\alpha r = \lim_{k \rightarrow \infty} \alpha \frac{|a_k|}{|a_{k+1}|}$, there is a $K \in \mathbb{N}$ such that

$$|z - z_0| < \alpha \frac{|a_k|}{|a_{k+1}|}$$

for all $k \geq K$.

- (b) For such k , $\frac{|a_{k+1}|}{|a_k|} |z - z_0| < \alpha$ and so $\lim_{k \rightarrow \infty} \frac{|a_{k+1}(z - z_0)^{k+1}|}{|a_k(z - z_0)^k|} \leq \alpha$.

- (c) Since $\alpha < 1$, it follows that

$$\lim_{k \rightarrow \infty} \frac{|a_{k+1}(z - z_0)^{k+1}|}{|a_k(z - z_0)^k|} < 1.$$

The Ratio Test says that $\sum_{k=0}^{\infty} |a_k(z - z_0)^k|$ converges. It follows that $\sum_{k=0}^{\infty} a_k(z - z_0)^k$ is absolutely convergent and therefore convergent.

- (d) Since the series converges for all z in the disc $\{z \in \mathbb{C} : |z - z_0| < r\}$, this disc is contained in the domain of convergence. Thus $r \leq R$.

- (e) Suppose now that $|z - z_0| > r$. This means that

$$\frac{|a_k|}{|a_{k+1}|} < |z - z_0|$$

for all sufficiently large $k \in \mathbb{N}$ and so

$$|a_k(z - z_0)^k| < |a_{k+1}(z - z_0)^{k+1}|$$

for all such k . But this implies that the terms of the series $\sum_{k=0}^{\infty} a_k(z - z_0)^k$ don't tend to 0, so the series is divergent. Since we have shown that the series diverges for all z outside the circle $\{z \in \mathbb{C} : |z - z_0| = r\}$, we can't have $R > r$.

2.(a) Yes: $f'(2) = \lim_{z \rightarrow 2} \frac{|z^2 - 4|^2 - 0}{z - 2} = \lim_{z \rightarrow 2} \frac{|z + 2|^2 |z - 2|^2}{z - 2} = \lim_{z \rightarrow 2} \left(|z + 2|^2 \frac{|z - 2|}{z - 2} |z - 2| \right) = 0.$

- (b) Yes, since f is differentiable.

- 3.(a) The Cauchy-Hadamard formula says that

$$\frac{1}{R} = \limsup_k (3 + (-1)^k) = 4.$$

Thus $R = \frac{1}{4}$.

(b) By the ratio formula,

$$R = \lim_{k \rightarrow \infty} \frac{k^k}{k!} \cdot \frac{(k+1)!}{(k+1)^{k+1}} = \lim_{k \rightarrow \infty} \left(\frac{k}{k+1} \right)^k = \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k} \right)^{-k} = e^{-1}.$$

4. We shall use the formula for a geometric progression:

$$\frac{1}{1-w} = 1 + w + w^2 + w^3 + \dots = \sum_{k=0}^{\infty} w^k.$$

This converges if and only if $|w| < 1$.

(a) There are several ways of tackling this one.

$$\begin{aligned} \frac{2+z}{1-iz} &= (2+z) \cdot \frac{1}{1-iz} \\ &= (2+z) \sum_{k=0}^{\infty} (iz)^k \quad (\text{putting } w = iz) \\ &= 2 \sum_{k=0}^{\infty} (iz)^k + z \sum_{k=0}^{\infty} (iz)^k \\ &= 2 \sum_{k=0}^{\infty} (iz)^k - i \sum_{k=0}^{\infty} (iz)^{k+1} \\ &= 2 + \sum_{k=1}^{\infty} (2-i) i^k z^k \\ &= 2 + \sum_{k=1}^{\infty} a_k z^k, \end{aligned}$$

where

$$a_k = \begin{cases} 2-i & \text{if } k \equiv 0 \pmod{4}, \\ 2i+1 & \text{if } k \equiv 1 \pmod{4}, \\ -2+i & \text{if } k \equiv 2 \pmod{4}, \\ -2i-1 & \text{if } k \equiv 3 \pmod{4}. \end{cases}$$

By the above, this converges $\iff |w| < 1 \iff |z| < 1$.

(b) If $z \neq \frac{1}{2}$, then $\frac{1}{4z^2 + 2z + 1} = \frac{1-2z}{1-8z^3}$. Thus, using $w = 8z^3 = (2z)^3$,

$$\begin{aligned} \frac{1}{4z^2 + 2z + 1} &= (1-2z) \sum_{k=0}^{\infty} (2z)^{3k} \\ &= \sum_{k=0}^{\infty} (2z)^{3k} - \sum_{k=0}^{\infty} (2z)^{3k+1} \\ &= \sum_{k=0}^{\infty} b_k z^k, \end{aligned}$$

$$\text{where } b_k = \begin{cases} 2^k & \text{if } k \equiv 0 \pmod{3}, \\ -2^k & \text{if } k \equiv 1 \pmod{3}, \\ 0 & \text{if } k \equiv 2 \pmod{3}. \end{cases}$$

The series converges $\iff |w| < 1 \iff |(2z)^3| < 1 \iff |z| < \frac{1}{2}$.

5.(a) (This is Q2(b) on Tut 5.) We split the path into pieces. The line segment joining 0 and 2 can be parametrised by $\gamma_1(t) = t$, where $0 \leq t \leq 2$, so

$$\int_{\gamma_1} \bar{z} dz = \int_0^2 t dt = 2.$$

The quartercircle can be parametrised by $\gamma_2(t) = 2e^{it}$, where $0 \leq t \leq \frac{\pi}{2}$. This yields

$$\begin{aligned} \int_{\gamma_2} \bar{z} dz &= \int_0^{\pi/2} 2e^{-it} \cdot 2ie^{it} dt \\ &= 2i\pi. \end{aligned}$$

The line segment joining $2i$ and 0 can be parametrised by $\gamma_3(t) = i(2 - t)$, where $0 \leq t \leq 2$, so

$$\int_{\gamma_3} \bar{z} dz = \int_0^2 (-i)(2 - t) \cdot (-i) dt = -2.$$

$$\text{Thus } \int_{\gamma} \bar{z} dz = 2i\pi.$$

(b) It is possible to evaluate this integral by paramtrising the path. However, it's quicker to note that $F(z) = -\frac{1}{2(z^2 - 4)}$ is an antiderivative of $f(z) = \frac{z}{(z^2 - 4)^2}$, so

$$\int_{\gamma} \frac{z}{(z^2 - 4)^2} dz = F(4) - F(0) = -\frac{1}{24} - \frac{1}{8} = -\frac{1}{6}.$$

6. γ traces out a semicircle $\mathcal{C}$ with $|\mathcal{C}| = 3\pi$. The Reverse Triangle Inequality says that

$$||z^4| - 16| \leq |z^4 - 16|$$

for all $z \in \mathbb{C}$. Thus

$$\left| \frac{1}{z^4 - 16} \right| \leq \frac{1}{|z|^4 - 16} = \frac{1}{81 - 16} = \frac{1}{65}$$

for all $z \in \mathcal{C}$ and so

$$\left| \int_{\gamma} \frac{1}{z^4 - 16} dz \right| \leq \frac{3\pi}{65}.$$